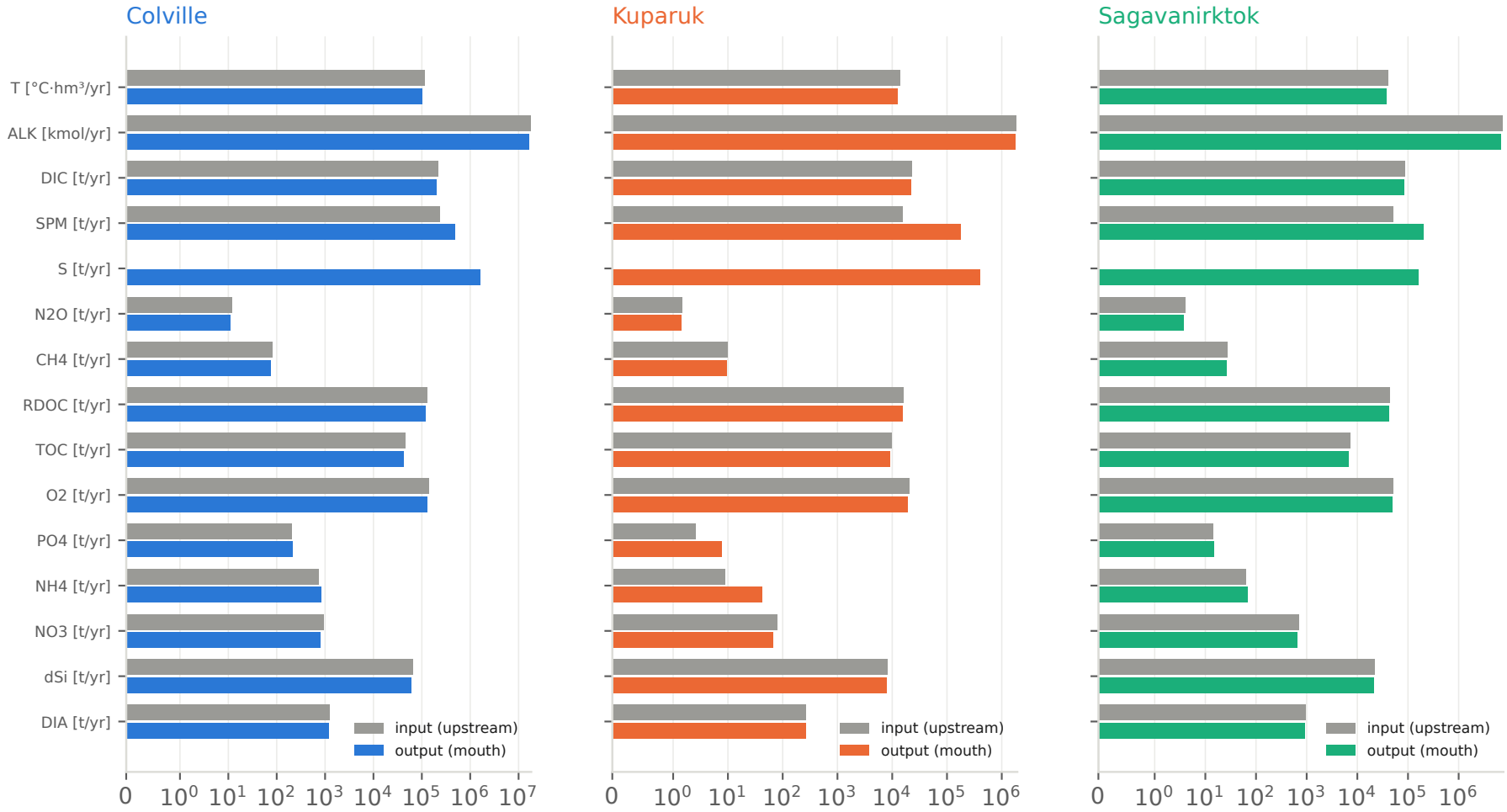
Estuarine filtering — annual load in vs. out (2006–2023 mean)



Input = upstream boundary, output = mouth, both 2 cells in from the boundary-forced cell, mean over all post-warmup years (2006–2023) of the 19-yr full-forcing record. Mouth loads are corrected for the ~12h-tide daily-snapshot aliasing bias (mouth_bias_correction -- calibrated once against runs/definitive_full/'s true 6-min output, real 2005-2006). Symlog x-axis (linear near 0). See filtering_mass_balance.csv for every year and the k=0 vs k=2 check. Upstream (river) boundaries are CONSTANT for every solute except T and TOC -- same limitation as the original 44-yr record. Marine (mouth) boundaries, by contrast, are genuinely time-varying here for EVERY solute (S, DIC, ALK, NO3, NH4, PO4, dSi, O2, TOC -- ECCO-Darwin v5, 2005-2023), which is new relative to the original figures (only T was marine-time-varying there). Structure seen here reflects BOTH the model's own internal dynamics acting on a constant upstream source AND a genuinely time-varying ocean signal propagating in from the mouth. See CLAUDE.md -> "Interannual forcing" -> "Full-forcing interannual (2005-2023)". This is an ADVECTIVE-flux-only estimate. For S (salinity) specifically, dispersive/tidal-pumping transport -- the dominant salt-intrusion mechanism on this coast (see CLAUDE.md -> "Width and dispersion") -- is NOT captured, so S's entry likely understates true salt exchange; the biogeochemical solutes are advection-dominated and less affected by this limitation.